PRODUCT VIABILITY

Viability Report

Value proposition, differentiation, adoption pathway and validation plan

VIAB-03

AAKAASH ADHINAYAK

VAYU NEXA AI | Alchemy Architects | SIH 2026

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1. Executive verdict

Technical feasibility asks whether AAKAASH ADHINAYAK can be built. Viability asks whether there is a credible user, repeatable value, adoption route and sustainable operating model. The strongest viable position is not “an AI dashboard for defence”; it is a provenance-aware planning and verification platform for training, logistics, maintenance and generic resource scheduling, with the defence-related scenario represented only as a non-operational student prototype unless authorised data and institutional approval are later obtained.

Evidence note: The public URL resolves to a page titled "AAKAASH ADHINAYAK - Verified Intelligence for Adaptive Resource Planning", but the available research interface did not expose navigable page text, runtime interactions, source code, API traces or test results. Therefore this report distinguishes observed deployment metadata and the team-supplied blueprint from proposed design and unverified implementation. It is not a source-code audit, security assessment, or certification of the live system.

2. Value proposition

Proposed value proposition: Give planning teams a reproducible way to combine records from multiple sources, expose data-quality and freshness problems, compare feasible schedules, independently verify hard constraints and generate an auditable explanation of each result.

Stakeholder	Potential value	Proof required
Student evaluators / SIH judges	Distinctive engineering demo with measurable solver behaviour, documented sources and reproducible tests.	Working end-to-end walkthrough, benchmark protocol and honest limitations.
Academic labs / training centres	Scenario-based teaching, data quality labs, scheduling experiments and reporting.	Usability sessions, learning outcomes and low-friction setup.
Civilian aviation or maintenance teams	Potential value from planning, source freshness, inspection scheduling and predictive-maintenance analysis.	Domain fit, dataset rights, process metrics and safety review.
Public-sector / defence-adjacent programmes	Potential sandbox for non-operational logistics, evidence traceability or research.	Procurement eligibility, authorised datasets, hosting/security requirements and sponsor validation.
Small maintenance/logistics organisations	Potential fit for generic resource and maintenance scheduling where current tools are fragmented.	Interviews, time-saved baseline, integration costs and willingness to pay.

3. Competitive position and defensibility

The category is crowded with dashboards, maintenance systems, operations research solvers and LLM copilots. The application should not claim that no similar product exists. A defensible differentiation is an implemented method of tying source provenance and freshness to a deterministic optimisation run, an independent validator, an explainable evidence graph and a reproducible benchmark. Individual components are not unique; the implementation quality, workflows, evaluation data and results create the proof of differentiation.

Alternative	Strength	Why AAKAASH can differentiate
Spreadsheet/manual scheduling	Accessible, flexible and familiar.	Repeatable constraint checks, provenance, replayable scenarios and audit snapshots.
General dashboards/BI	Strong trend reporting and visualisation.	Candidate-plan generation plus hard-constraint verification.
Standalone optimisation solver	Efficient mathematical solution search.	Source-quality context, independent post-checks, human review and evidence export.
Generic LLM copilot	Natural-language explanation and summarisation.	Tool-restricted access to the actual run and proof that output respects data state.

Alternative	Strength	How AAKAASH can differ
Enterprise maintenance/ERP tools	Deep domain records, mature integrations and support.	A small, transparent research sandbox; not a claimed replacement for certified enterprise systems.

4. Adoption path

Stage 1 - Student demonstrator

Use open/public datasets and synthetic generic schedule records, create reproducible tests, document data lineage and publish a non-sensitive demo. This stage validates engineering quality, not market demand.

Stage 2 - Academic or research pilot

Run structured sessions with faculty, logistics researchers or maintenance educators. Compare the current process with the prototype on the same dataset and task set. Measure input preparation time, constraint errors detected, schedule quality, run reproducibility and ease of understanding. Collect feedback without collecting unnecessary personal or operational information.

Stage 3 - Civilian-domain pilot

Pick a bounded civilian use case such as generic maintenance intervals, inventory replenishment or resource scheduling. Establish permitted data flows, a named sponsor, integration requirements, information-security review and a baseline before claiming efficiency gains.

Stage 4 - Institutional adoption assessment

For any sensitive or public-sector deployment, require formal sponsorship, appropriate approval, procurement review, hosting/security assessment, governance ownership and authorised datasets. Do not imply that an SIH prototype or public web URL constitutes MoD endorsement.

5. Validation of user need and willingness to pay

- Conduct 8-12 exploratory interviews across likely civilian/academic user groups; treat them as research, not a representative market survey.
- Map each pain point to a user action and observable metric; avoid broad “AI saves money” statements without baseline data.
- Run at least three scenario-based usability tests with users who did not build the app.
- Ask prospective pilot users to rank value, integration burden, required security controls and acceptable support model.
- Request a concrete next step: data access, a pilot sponsor, a letter of interest, or a paid discovery engagement. Verbal enthusiasm alone is weak evidence.
- Record negative outcomes and reasons; do not report hypothetical customers as customers.

6. Product viability KPIs

KPI	Definition	Evidence source
Active pilot users	Users who complete a defined workflow within a period.	Product analytics with consent and privacy controls.
Workflow completion	Share of test scenarios completed without facilitator intervention.	Observed usability sessions and event logs.
Constraint violation escape rate	Invalid candidate schedules incorrectly marked validated divided by invalid test schedules. Target 0 in the defined acceptance suite.	Mutation tests and independent-validator suite.
Decision traceability	Share of sampled displayed outputs that link back to inputs, source versions and validation result.	Evidence-ledger audit.
Data integration time	Elapsed person-time to add a documented supported dataset.	Timed integration protocol.
Value proof	Change in elapsed planning time or error rate against a pre-defined baseline.	Paired, repeatable experiments.
Retention/renewal intent	Whether a pilot user returns, requests expanded use or provides budget-backed intent.	Pilot follow-up and procurement evidence.

7. Viability risks

Risk	Implication	Response
Problem is too broad	Users cannot identify the first workflow or outcome.	Choose one narrow pilot use case and define exclusions.
No authorised data	Demo may not transfer to a real environment.	Prove public-data analysis separately from generic synthetic solver testing.
Hard enterprise integration	Integration costs may exceed value for a small pilot.	CSV/JSON first; versioned adapters; add API/ERP connectors only after sponsor confirmation.
Public-sector purchasing cycle	Long sales cycle and high evidence requirements.	Validate through academia/civilian pilots first; treat public procurement as a separate track.
Unproven outcome improvement	Commercial promises lack credibility.	Publish baselines, measurement protocol, raw run results and limitations.

8. Viability conclusion

Decision: VIABLE AS A RESEARCH / EDUCATIONAL PROTOTYPE; COMMERCIAL VIABILITY NOT YET PROVEN. There is a credible pathway through an academic or civilian-domain pilot. Stronger viability evidence requires user interviews, a repeated workflow, benchmark results, a pilot sponsor and willingness-to-pay data. Do not position this as deployable defence software without the relevant approvals and assurance.

References: [R1], [R2], [R7], [R14]. Full URLs are available in the separate References Report.